

2. This question is about the Solar System.

(a) **Complete** the following sentence. [2]

The Solar System was formed by the collapse of a cloud of and
..... .

(b) The table shows some information about planets in our Solar System.

	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
mean distance from Sun (AU)	0.4	0.7	1	1.5	5.2	9.5	19.2	30
mass compared with Earth	0.055	0.815	1	0.107	318	95	15	17
orbital time (years)	0.24	0.60	1	2	12	30	84	160
mean temperature at surface on sunny side (°C)	430	465	20	−20	−150	−170	−200	−210
number of moons	0	0	1	2	63	60	27	13

Use the information in the table to answer the questions that follow.

(i) Describe how the temperature on a planet depends on the distance from the Sun. [1]

.....
.....

(ii) Identify the planet that does not follow this trend. [1]

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- (iii) Pallas is an asteroid found in the asteroid belt.
- I. Estimate its temperature °C [1]
- II. Estimate its orbital time years [1]
- (iv) It is suggested that the greater the mass of a planet, the more moons that will orbit around it. Explain whether the data agrees with this suggestion. [2]

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